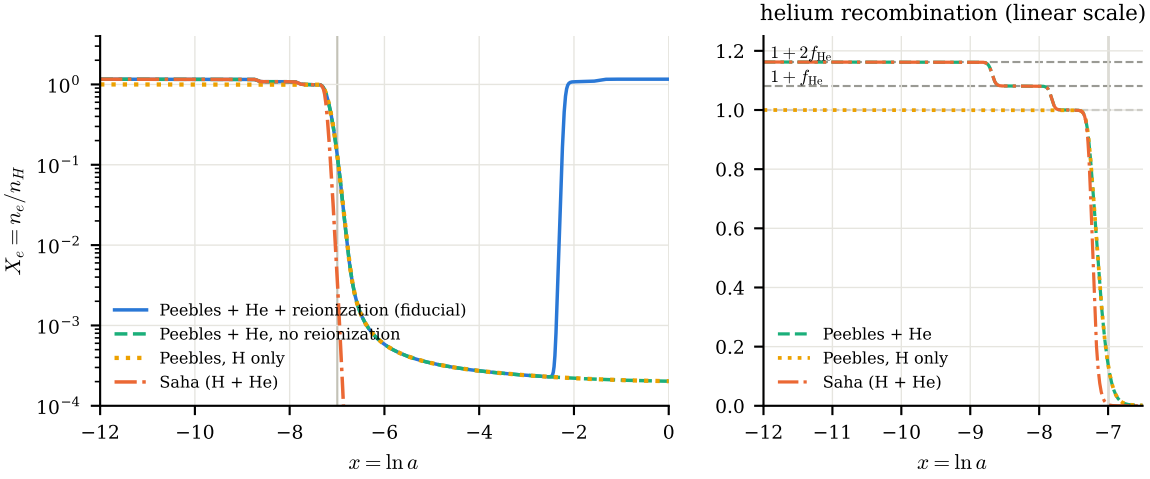
$X_e = n_e/n_H$
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Peebles + He + reionization (fiducial)
Peebles + He, no reionization
Peebles, H only
Saha (H + He)
−12
−10
−8
−6
−4
−2
0
$x = \ln a$
helium recombination (linear scale)
$1 + 2f_{\rm He}$
$1 + f_{\rm He}$
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Peebles + He
Peebles, H only
Saha (H + He)
−12
−11
−10
−9
−8
−7
$x = \ln a$